INCH-POUND

MS25030M
28 April 1995
SUPERSEDING
MS25030L
31 August 1982

MILITARY SPECIFICATION SHEET

RELAY, 50 AMP, 2 PDT N.O. TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED

(M) Inactive for new design effective 28 April 1995.
No superseding standard.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

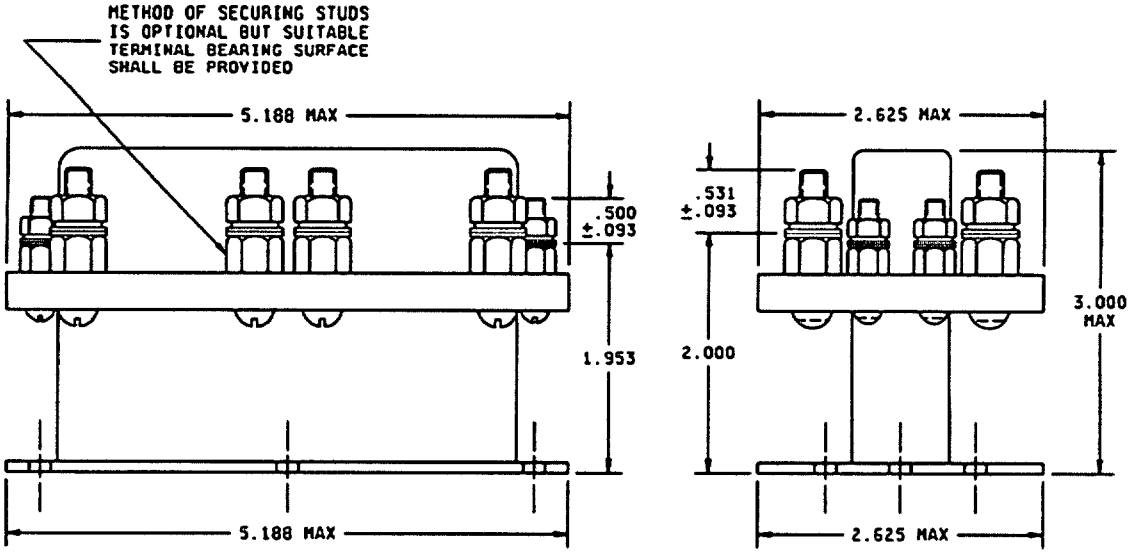
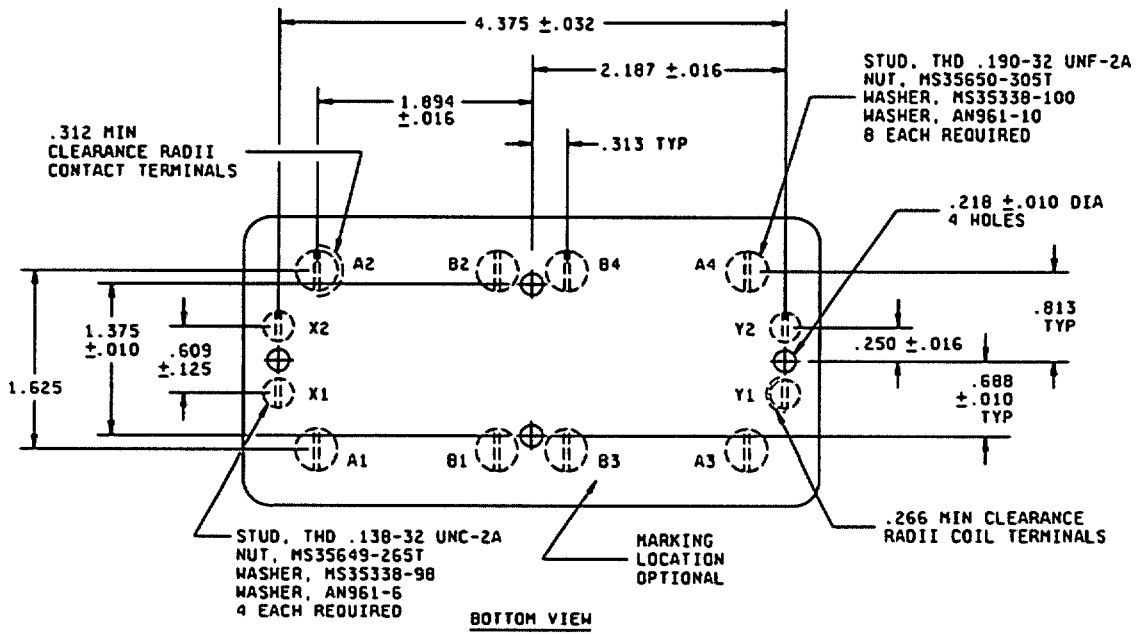


FIGURE 1. Dimensions and configuration.

(M) denotes changes

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Inches	mm	Inches	mm
.010	0.25	.688	17.48
.093	2.36	.813	20.65
.125	3.18	1.375	34.93
.138	3.51	1.625	41.28
.190	4.83	1.938	49.23
.218	5.54	1.953	49.61
.266	6.76	2.000	50.80
.305	7.75	2.281	57.94
.312	7.93	2.625	66.68
.313	7.95	3.000	76.20
.500	12.70	4.562	115.88
.531	13.49	5.188	131.78
.609	15.47		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.062 (1.57 mm).
4. Coil and auxiliary terminals may use an additional flat washer for terminal seat.
5. Mechanical linkage shall be provided to prevent both relays from being closed simultaneously.
6. Referenced documents shall be of the issue in effect on the date of invitation for bid.
7. This standard takes precedence over the procurement documents referenced herein.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 400 amperes.

Rupture current (NO): 500 amperes.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case:	1,250	N/A	1,000	N/A
Aux. contacts:	1,250	N/A	1,000	N/A
All other points:	1,250	1,500	1,000	1,125

Altitude, 1 minute:

	28 V dc	115 V ac
Coil to case:	500	N/A
Aux. contacts:	500	N/A
All other points:	500	500

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Environmental characteristics:

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet.

Shock, g-level: 25 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration, sinusoidal: See table IV.

Acceleration: 10 g's.

Part or Identifying Number (PIN): MS25030- (plus applicable dash number from table I).

Qualification by similarity: See table V.

(M) Quality assurance provisions: Group B and group C testing are not required. In the event of a change in the design or construction of the part, the manufacturer shall notify the qualifying activity and shall impose additional testing requirements as necessary.

(M) Supersession data: See table VI.

TABLE I. Dash numbers and general characteristics. 1/ 2/

PIN	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary contacts	Max weight (pounds) 1/
MS25030-D1B	II	dc	Stud	Plate	N/A	1.7

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TABLE II. Rated contact load (amperes per pole) (case grounded). 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase				See notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	50				50								
Inductive	10	50												
Motor	50	50				35								
Lamp														
Transfer load														2/
Mechanical life (reduced current)	100	12.5				12.5								
Minimum current	50	5												

1/ Absence of value indicates that parameter is not applicable to this specification sheet.
2/ Transfer load indicates that relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

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TABLE III. Operating characteristics.

PIN MS25030-	Coil data												Time (milliseconds maximum) 2/					
	Coil	Rated			Max		Max pick-up voltage 1/			Hold voltage 1/	Drop-out voltage 1/		Operate	Release	Bounce			
															Main		Aux	
		Volts	Freq. Hz	Res Ω +15% -10%	Volts	A	Normal 1/	High temp test	Cont current test						NO	NC	NO	NC
D1B	X1,X2 Y1,Y2	28	DC	66	29	.5	18	21	22.5	7.0	1.5		25	15	5	n/a	n/a	n/a

1/ Over the temperature range.
2/ With rated coil voltage.

TABLE IV. Vibration levels (sinusoidal).

PIN MS25030-	Frequency				
	5 Hz-10 Hz	10 Hz-55 Hz	55 Hz-250 Hz	250 Hz-500 Hz	500 Hz-1500 Hz
D1B	.08 DA	.06 DA	2 g's	2 g's	n/a

TABLE V. Qualification by similarity. 1/

PIN MS25030-	Loads						Dynamics 2/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	X	Y	A	B	C
D1B	4						2			4		

1/ Absence of value indicates that parameter is not applicable.
2/ All units must be tested. (Reference amendment 1, appendix 1, MIL-R-6106, construction, internal).

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(M) TABLE VI. Supersession data.

Superseded PIN MS25030-	Replacement PIN MS25030-
2	D1B
3	D1B
D1A	D1B

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 85

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0945)

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